

Data modeling: CSCI E-106

Polynomial Regressions

Uses Polynomial Models

- Polynomial regression models for quantitative predictor variables
 - the most frequently used curvilinear response models
 - handled easily
 - special case of the general LRM

Uses Polynomial Models, cont'd

- Two basic types
 - The true curvilinear response function is a polynomial function
 - The true curvilinear response function is **unknown**; a polynomial function is a **good approximation to the true function**
- Polynomial regression models may provide good fits for the data at hand, but may turn in unexpected directions when extrapolated beyond the range of the data

Polynomial Regression Models

- Polynomial regression models may contain one, two, or more than two predictor variables.
- Each predictor variable may be present in various powers

A second order model with one predictor variable

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i$$

Where $x_i = (X_i - \bar{X})$.

- Predictor variable is centered ($x_i = (X_i - \bar{X})$) : X and X^2 will be highly correlated.
- Centering the predictor variable reduces the multicollinearity

Second order with predictor variable

- The regression coefficients in polynomial regression are frequently written in a slightly different fashion, to reflect the pattern of the exponents:

$$Y_i = \beta_0 + \beta_1 x_i + \beta_{11} x_i^2 + \varepsilon_i, \text{ where } x_i = (X_i - \bar{X}).$$

- The quadratic response function: parabola $\rightarrow E\{Y\} = \beta_0 + \beta_1 x + \beta_{11} x^2$

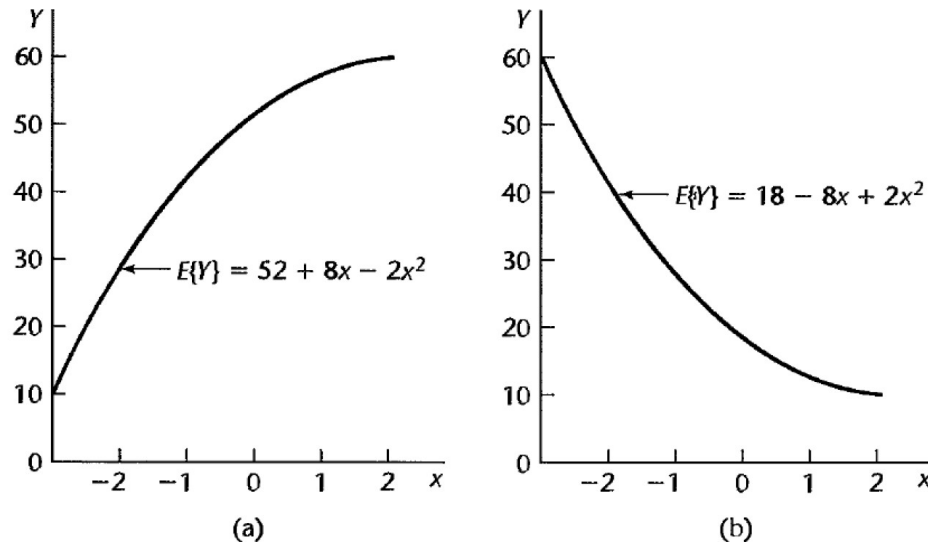


Figure : Examples of Second-Order Polynomial Response Functions.

Second order with predictor variable, cont'd

$$E\{Y\} = \beta_o + \beta_1 x + \beta_{11} x^2$$

Parameters:

- β_o : the mean response of Y when $x = 0$ (i. e $X = \bar{X}$)
- β_1 : the linear effect coefficient
- β_{11} : the quadratic effect coefficient

Third order with predictor variable

$$Y_i = \beta_0 + \beta_1 x_i + \beta_{11} x_i^2 + \beta_{111} x_i^3 + \varepsilon_i$$

where $x_i = (X_i - \bar{X})$.

The response function for regression model:

$$E\{Y\} = \beta_0 + \beta_1 x + \beta_{11} x^2 + \beta_{111} x^3$$

Higher orders with one predictor variable

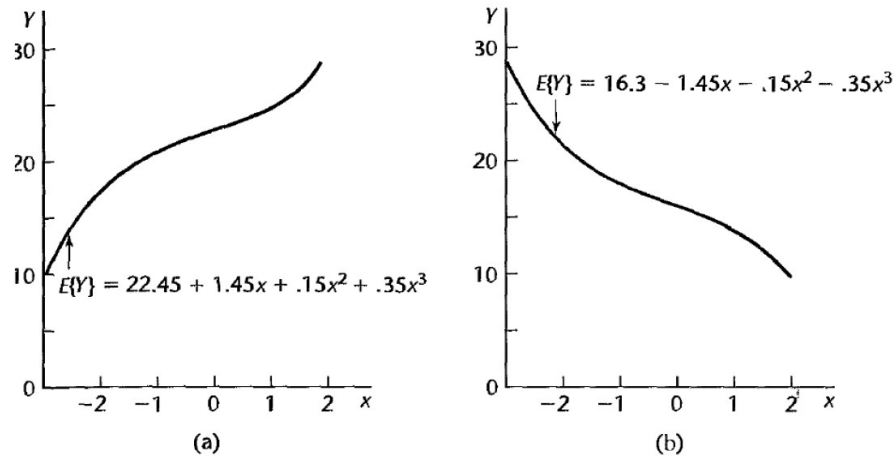


Figure : Examples of Third-Order Polynomial Response Functions.

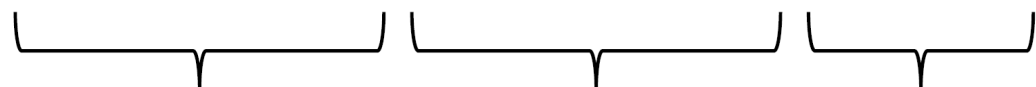
- Employed with special caution
- The **interpretation** of the coefficients becomes difficult
- The models may be highly erratic for interpolations and small extrapolations
- A polynomial model of sufficiently high order can always be found to fit data containing no repeat observations perfectly.
 - Illustration: the fitted polynomial regression function for one predictor variable of order $n - 1$ will pass through all n observed Y values.

Second order with two predictor variables

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_{11} x_{i1}^2 + \beta_{22} x_{i2}^2 + \beta_{12} x_{i1} x_{i2} + \varepsilon_i ,$$

where $x_{i1} = (X_{i1} - \bar{X}_1)$, and $x_{i2} = (X_{i2} - \bar{X}_2)$

- The response function: conic section

$$E\{Y\} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2$$


The diagram shows three horizontal curly braces under the equation above. The first brace is under $\beta_1 x_1 + \beta_2 x_2$, the second is under $\beta_{11} x_1^2 + \beta_{22} x_2^2$, and the third is under $\beta_{12} x_1 x_2$. Each brace has a vertical line pointing down to its corresponding label below.

Linear component Quadratic component Cross product

- β_{12} : the interaction effect coefficient
- The second-order model with three predictor variables is similar

Hierarchical Approach to Fitting

- Special cases of the general linear regression model
- Fitting of polynomial models: no new problems
- All earlier results on fitting apply on making inferences.

Implementation of Polynomial Regression Models

Ideas to find an approximation to the true regression function:

- 1 often fit a **second-order** or **third-order** model

$$Y_i = \beta_0 + \beta_1 x_i + \beta_{11} x_i^2 + \beta_{111} x_i^3 + \varepsilon_i$$

- 2 explore whether a **lower-order** model is adequate
 - Test $\beta_{111} = 0$
 - Or test not both $\beta_{11} = 0; \beta_{111} = 0$
- 3 The decomposition of SSR into extra sums of squares:

$$SSR(x); \quad SSR(x^2|x); \quad SSR(x^3|x, x^2)$$

- 4 Test whether $\beta_{111} = 0$: $SSR(x^3|x, x^2)$
Test whether $\beta_{11} = \beta_{111} = 0$:
 $SSR(x^2, x^3|x) = SSR(x^2|x) + SSR(x^3|x, x^2)$

Implementation of Polynomial Regression Models, cont'd

- If a polynomial term of a given order is retained, then all related terms of lower order are also retained in the model.
 - providing more basic information about the shape of the response function

- Wish to express the model in terms of the original variables:

$$\begin{aligned}\hat{Y} &= b_0 + b_1x + b_{11}x^2 \\ \hat{Y} &= b_0 + b_1(X - \bar{X}) + b_{11}(X - \bar{X})^2 \\ \hat{Y} &= (b_0 - b_1\bar{X} + b_{11}\bar{X}^2) + (b_1 - 2b_{11}\bar{X})X + b_{11}X^2 \\ \hat{Y} &= b'_0 + b'_1X + b'_{11}X^2\end{aligned}$$

- The fitted values and residuals for the regression function in terms of x or X are the same.
- Centered observations: to reduce potential calculation difficulties; multicollinearity;

Case Example: the life of power cell

- Studied the effects of the charge rate and temperature on the
- life of a new type of power cell
- X_1 : the charge rate at three level (0.6, 1.0, 1.4 amperes)
- X_2 : the ambient temperature at three levels (10, 20, 300C)
- Y : the life of the power cell
- don't know the nature of the response function in the range of the factors studied
- fit the second-order polynomial regression model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_{11} x_{i1}^2 + \beta_{22} x_{i2}^2 + \beta_{12} x_{i1} x_{i2} + \varepsilon_i ,$$

$\Rightarrow$

$$E\{Y\} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2$$

Case Example: the life of power cell, cont'd

Table : Data-Power Cells Example.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Number of Cycles	Charge Rate	Temperature	Coded	Values			
Cell	(thousands)	(thousands)						
i	Y_i	X_{i1}	X_{i2}	x_{i1}	x_{i2}	x_{i1}^2	x_{i2}^2	$x_{i1}x_{i2}$
1	150	0.6	10	-1	-1	1	1	1
2	86	1.0	10	0	-1	0	1	0
3	49	1.4	10	1	-1	1	1	-1
4	288	0.6	20	-1	0	1	0	0
5	157	1.0	20	0	0	0	0	0
6	131	1.0	20	0	0	0	0	0
7	184	1.0	20	0	0	0	0	0
8	109	1.4	20	1	0	1	0	0
9	279	0.6	30	-1	1	1	1	-1
10	235	1.0	30	0	1	0	1	0
11	224	1.4	30	1	1	1	1	1
		$\bar{X}_1 = 1.0$	$\bar{X}_2 = 20$					

$$x_{i1} = \frac{X_{i1} - \bar{X}_1}{0.4}; \quad x_{i2} = \frac{X_{i2} - \bar{X}_2}{10}$$

Case Example: the life of power cell, cont'd

```
Power.Cell <- read.csv("/cloud/project/Data/Power Cell.csv")
attach(Power.Cell )
X1sqr<-X1^2
X2sqr<-X2^2
X1X2<-X1*X2
x1<-(ex$X1-mean(ex$X1))/0.4
x2<-(ex$X2-mean(ex$X2))/10
x1sqr<-x1^2
x2sqr<-x2^2
x1x2<-x1*x2
f<-lm(Y~ x1+ x2+ x1sqr+ x2sqr+ x1x2)
```

Case Example: the life of power cell, cont'd

Correlation between		Correlation between	
X_1 and X_1^2 :	.991	X_2 and X_2^2 :	.986
x_1 and x_1^2 :	0.0	x_2 and x_2^2 :	0.0

```
> round(cor(cbind(X1,X2,X1sqr,X2sqr,X1X2)),4)
      X1      X2 X1sqr X2sqr X1X2
X1  1.0000 0.0000 0.9910 0.0000 0.6053
X2  0.0000 1.0000 0.0000 0.9861 0.7566
X1sqr 0.9910 0.0000 1.0000 0.0059 0.5999
X2sqr 0.0000 0.9861 0.0059 1.0000 0.7461
X1X2  0.6053 0.7566 0.5999 0.7461 1.0000
```

```
> round(cor(cbind(x1,x2,x1sqr,x2sqr,x1x2)),4)
      x1 x2 x1sqr x2sqr x1x2
x1  1  0 0.0000 0.0000  0
x2  0  1 0.0000 0.0000  0
x1sqr 0  0 1.0000 0.2667  0
x2sqr 0  0 0.2667 1.0000  0
x1x2  0  0 0.0000 0.0000  1
```

- Interested in whether **interaction** effects and **curvature** effects are required in the model
- fitted model:

$$\hat{Y} = 162.84 - 55.83x_1 + 75.50x_2 + 27.39x_1^2 - 10.61x_2^2 + 11.50x_1x_2$$

Case Example: the life of power cell, cont'd

```
> summary(lm(Y~x1+x2+x1sqr+x2sqr+x1x2))
```

Call:

```
lm(formula = Y ~ x1 + x2 + x1sqr + x2sqr + x1x2)
```

Residuals:

1	2	3	4	5	6	7	8	9
-21.465	9.263	12.202	41.930	-5.842	-31.842	21.158	-25.404	-20.465
10	11							
7.263	13.202							

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	162.84	16.61	9.805	0.000188	***
x1	-55.83	13.22	-4.224	0.008292	**
x2	75.50	13.22	5.712	0.002297	**
x1sqr	27.39	20.34	1.347	0.235856	
x2sqr	-10.61	20.34	-0.521	0.624352	
x1x2	11.50	16.19	0.710	0.509184	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 32.37 on 5 degrees of freedom

Multiple R-squared: 0.9135, Adjusted R-squared: 0.8271

F-statistic: 10.57 on 5 and 5 DF, p-value: 0.01086

Case Example: the life of power cell, cont'd

Model: MODEL1

Dependent Variable: Y

Analysis of Variance

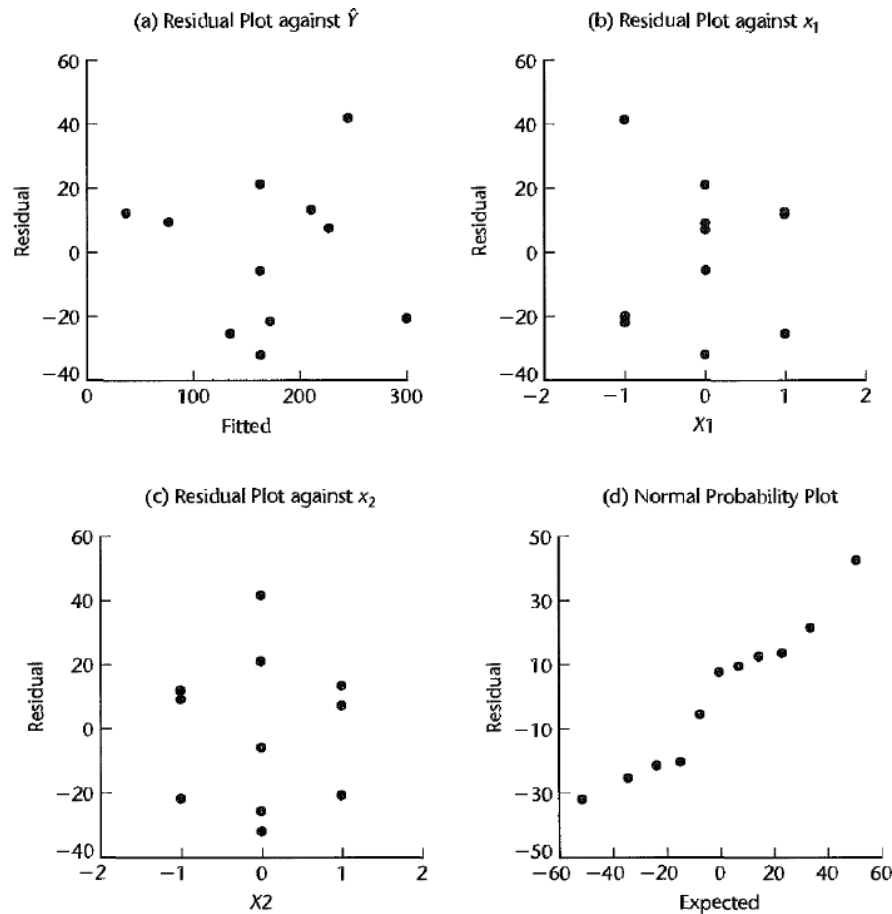
Source	DF	Sum of Squares	Mean Square	F Value	Prob>F
Model	5	55365.56140	11073.11228	10.565	0.0109
Error	5	5240.43860	1048.08772		
C Total	10	60606.00000			
Root MSE		32.37418	R-square	0.9135	
Dep Mean		172.00000	Adj R-sq	0.8271	
C.V.		18.82220			

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	T for H0: Parameter=0	Prob > T
INTERCEP	1	162.842105	16.60760542	9.805	0.0002
X1	1	-55.833333	13.21670483	-4.224	0.0083
X2	1	75.500000	13.21670483	5.712	0.0023
X1SQ	1	27.394737	20.34007956	1.347	0.2359
X2SQ	1	-10.605263	20.34007956	-0.521	0.6244
X1X2	1	11.500000	16.18709146	0.710	0.5092

Variable	DF	Type I SS
INTERCEP	1	325424
X1	1	18704
X2	1	34202
X1SQ	1	1645.966667
X2SQ	1	284.928070
X1X2	1	529.000000

Case Example: the life of power cell, cont'd



- None of the plots suggest any gross inadequacies of the model
- The coefficient of correlation between the ordered residuals and their expected values: **0.974**

Figure : Diagnostic Residual Plots-Power Cells Example.

Case Example: the life of power cell, cont'd

- **Test of fit:** replications at $x_1 = 0, x_2 = 0$

$$SSPE = 1, 404.67 \quad (df = n - c = 2)$$

$$SSLF = 3, 835.77 \quad (df = c - p = 3)$$

$$F^* = 1.82 \leq F(0.95; 3, 2) = 19.2$$

conclude that the second-order polynomial regression function is a good fit

- **Coefficient of Multiple Determination:** $R^2 = \%91$ and $R^2_{adj} = \%83$

The variation in the lives of the power cells is reduced by about 91% when the first-order and second-order relations to the change rate and ambient temperature are utilized.

Case Example: the life of power cell, cont'd

Multiple regression: lack of fit

Method 1

```
fit<-lm(Y~x1+x2+x1sqr+x2sqr+x1x2)
```

```
exfactor = factor( c(seq(-4,-1), rep(0,3),seq(1,4)) )
```

```
#fit full model
```

```
anova( fit, lm(Y ~ exfactor))
```

Method 2

```
library(alr3) # for lack of fit
```

```
pureErrorAnova(fit)
```

Case Example: the life of power cell, cont'd

Analysis of Variance Table

Response: Y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
x1	1	18704	18704	26.6315	0.03556	*
x2	1	34202	34202	48.6970	0.01992	*
x1sqr	1	1646	1646	2.3436	0.26546	
x2sqr	1	285	285	0.4057	0.58935	
x1x2	1	529	529	0.7532	0.47696	
Residuals	5	5240	1048			
Lack of fit	3	3836	1279	1.8205	0.37378	
Pure Error	2	1405	702			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Case Example: the life of power cell, cont'd

Partial F Test: whether a first-order model would be sufficient?

$$H_0: \beta_{11} = \beta_{22} = \beta_{12}$$

H_a : not all β s in H_0 equal to zero

$$\Rightarrow F^* = \frac{SSR(x_1^2, x_2^2, x_1x_2 | x_1, x_2)}{3} \div MSE = \frac{2,459.9}{3} \div 1048.1$$

$$= 0.78 \leq F(0.95; 3, 5) = 5.41$$

$\Rightarrow$ conclude H_0 : **no curvature and interaction effects** are needed

```
> anova( lm(Y~x1+x2), fit)
Analysis of Variance Table
Model 1: Y ~ x1 + x2
Model 2: Y ~ x1 + x2 + x1sqr + x2sqr + x1x2
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1      8 7700.3
2      5 5240.4  3    2459.9 0.7823 0.5527
```

Case Example: the life of power cell, cont'd

First-order Model:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

$$\Rightarrow \hat{Y} = 172.00 - \underset{(12.67)}{55.83}x_1 + \underset{(12.67)}{75.50}x_2$$

$$\Rightarrow \hat{Y} = 160.58 - 139.58X_1 + 7.55X_2 \quad \text{the original variables}$$

Case Example: the life of power cell, cont'd

- Estimation of Regression Coefficients:
to estimate the linear effect of the two predictor variables with a 90% family confidence coefficients by the Bonferroni method.
($g = 2$)

$$B = t\left(1 - 0.1/2(2)\right) = 2.306$$

$$s\{b'_1\} = \left(\frac{1}{0.4}\right)s\{b_1\} = 31.68$$

$$s\{b'_2\} = \left(\frac{1}{10}\right)s\{b_2\} = 1.267$$

$$\Rightarrow -212.6 \leq \beta_1 \leq -66.5 \quad 4.6 \leq \beta_2 \leq 10.5$$

Case Example: the life of power cell, cont'd

$$-212.6 \leq \beta_1 \leq -66.5 \quad 4.6 \leq \beta_2 \leq 10.5$$

- With confidence 0.90, we conclude that the mean number of charge/discharge cycles before failure decreases by 66.5 to 212.6 cycles with a unit increase in the charge rate for given ambient temperature, and increase by 4.6 to 10.5 cycles with a unit increase of ambient temperature for given charge rate.

```
> g = length(coef(lm(Y ~ X1+X2))) - 1
> alpha=0.1
> confint( lm(Y ~ X1+X2), level = 1-(alpha/g) )
              2.5 %      97.5 %
(Intercept)  64.617930 256.54874
X1           -212.602056 -66.56461
X2             4.629251  10.47075
```

Some further Comments on polynomial regression

Drawbacks: polynomial models

- Such models can be more expensive in degrees of freedom than alternative nonlinear models or linear models with transformed variables.
- serious multicollinearity may be present even when the predictor variables are centered.
- An alternative to using centered variables is to use orthogonal polynomials. Orthogonal polynomials are uncorrelated. (Some computer packages used orthogonal polynomials).
- Sometimes a quadratic response function is fitted for the purpose of establishing the linearity of the response function when repeat observations are not available.

Interaction Regression Models

Interaction effects:

- A regression model with $p - 1$ predictor variables contains additive effects:

$$E\{Y\} = f_1(X_1) + f_2(X_2) + \dots + f_{p-1}(X_{p-1})$$

- $f_i, i = 1, \dots, p - 1$: any functions, not necessarily simple one
- Illustration:

$$E\{Y\} = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_1^2}_{f_1(X_1)} + \underbrace{\beta_3 X_2}_{f_2(X_2)}$$

$\Rightarrow$ the effects of X_1, X_2 on Y are **additive**

$$E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 \quad (\times)$$

Interaction Regression Models, cont'd

- a cross-product term: modelling the interaction effect of two predictor variables on the response variable; ex: $\beta_3 X_1 X_2$
- The cross-product term: called an interaction term; a linear-by-linear or a bilinear interaction term

Interaction Regression Models, cont'd

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} X_{i2} + \varepsilon_i$$

- The meaning of β_1, β_2 is not the same as that given earlier ($\because \beta_3 X_{i1} X_{i2}$)
- β_1, β_2 no longer indicate the change in the mean response with a unit increase of the predictor variable with the other predictor variable held constant.
- the change in the mean response with a unit increase in X_1 when X_2 is held constant:

$$\frac{\partial E\{Y\}}{\partial X_1} = \beta_1 + \beta_3 X_2 \quad (\text{depend on } X_2)$$

Implementation of interaction regression models

Considerations:

- high multicollinearities: center the predictor variables

$$x_{ik} = X_{ik} - \bar{X}_{ik}$$

- The number of predictor variables is large $\Rightarrow$ the potential number of interaction terms becomes very large
- utilizing a priori knowledge
- plot the residuals for the additive regression model vs. the different interaction terms to determine which ones appear to be influential

Implementation of interaction regression models, cont'd

Body fat example:

- Model: Three predictor variables

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i1} X_{i2} + \beta_5 X_{i1} X_{i3} + \beta_6 X_{i2} X_{i3} + \varepsilon_i$$

- Some of the predictor variables are highly correlated with some of the interaction terms
- Centered variables:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i1} x_{i2} + \beta_5 x_{i1} x_{i3} + \beta_6 x_{i2} x_{i3} + \varepsilon_i$$

⇒

$$\hat{Y} = 20.53 + 3.438x_1 - 2.095x_2 - 1.616x_3 + .00888x_1x_2 - .08479x_1x_3 + .09042x_2x_3$$

$$MSE = 6.745$$

Implementation of interaction regression models, cont'd

Variable	Extra Sum of Squares
x_1	$SSR(x_1) = 352.270$
x_2	$SSR(x_2 x_1) = 33.169$
x_3	$SSR(x_3 x_1, x_2) = 11.546$
$x_1 x_2$	$SSR(x_1 x_2 x_1, x_2, x_3) = 1.496$
$x_1 x_3$	$SSR(x_1 x_3 x_1, x_2, x_3, x_1 x_2) = 2.704$
$x_2 x_3$	$SSR(x_2 x_3 x_1, x_2, x_3, x_1 x_2, x_1 x_3) = 6.515$

$$H_0: \beta_4 = \beta_5 = \beta_6$$

H_a : not all β s in H_0 equal to zero

$$\Rightarrow F^* = \frac{SSR(x_1 x_2, x_1 x_3, x_2 x_3 | x_1, x_2, x_3)}{3} \div MSE$$

$$= 0.53 \leq F(0.95; 3, 13) = 3.41$$

$\Rightarrow$ conclude H_0 : the interaction terms are not needed in the regression model. (P -value=0.67)